

2014

Question 15

_____ collection of nerve cell bodies in the CNS
and _____ collection of cell bodies in the PNS.

- A. Ganglion, Nucleus
- B. Nucleus, Ganglion
- C. Tract, Nerve
- D. Nerve, Tract

Correct Answer: B. Nucleus, Ganglion

Explanation:

✓ In the nervous system, a **nucleus** is defined as a collection of nerve cell bodies located **within the Central Nervous System (CNS)**.

✓ A **ganglion** is defined as a collection of nerve cell bodies located **outside the CNS** (within the Peripheral Nervous System, or PNS).

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 16

Which one of the following connective tissue covering is the thickest

- A. Epineurium
- B. Endoneurium
- C. Perineurium
- D. Fascicles

Correct Answer: A. Epineurium

Explanation:

◇ Peripheral nerves are wrapped in three concentric layers of protective connective tissue:

1, **Epineurium**: The **outermost, toughest, and thickest layer** of dense connective tissue that encloses the entire peripheral nerve trunk.

2, **Perineurium**: The middle layer that surrounds bundles of nerve fibers (fascicles).

3, **Endoneurium**: The thinnest, innermost layer of delicate connective tissue surrounding each individual myelinated or unmyelinated axon.

◇ *Fascicles* (Option D) are the bundles of nerve fibers themselves, not a protective tissue sheath.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 19

Which one of the following epidermal cell type has structural cell

- A. Melanocytes
- B. Langerhans cells
- C. Keratinocytes
- D. Merkel cells

Correct Answer: C. Keratinocytes

Explanation: The epidermis is defined as an avascular keratinized layer of stratified squamous epithelium.

✓ **Keratinocytes** are the primary structural cells of the epidermis, producing keratin to provide protective strength and structure to the skin.

✓ *Melanocytes* produce pigment, *Langerhans cells* function in immune defense, and *Merkel cells* serve as light-touch receptors.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 20

"Prickly layer" stands for which of the following epidermal stratum or layer

- A. Basale
- B. Spinosum
- C. Granulosum
- D. Lucidum
- E. Corneum

Correct Answer: B. Spinosum

Explanation: In histology and integumentary anatomy, the **Stratum Spinosum** is also known as the **"prickly layer"** (or spinous layer). This name comes from the spine-like or "prickly" appearance of its keratinocytes under microscopy, which is caused by the shrinkage of cells around their abundant desmosomal cell-to-cell connections.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 24

Endomysium covers

- A. muscle fiber
- B. muscle fascicle
- C. Tendon
- D. none

Correct Answer: A. muscle fiber

Explanation: According to the muscular organization section, the connective tissue coverings of a skeletal muscle are layered as follows:

◇ **Endomysium:** Encloses **each individual muscle fiber**.

◇ **Perimysium:** Surrounds smaller bundles of muscle fibers (muscle fascicles).

◇ **Epimysium:** Encloses the entire muscle.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 28

Which is not true

- A. Red blood cells have 120 days life span
- B. Red blood cells have biconcave disc shape
- C. Monocyte is categorized under granular leukocyte
- D. Platelets help for blood clotting

Correct Answer: C. Monocyte is categorized under granular leukocyte

Explanation: This statement is **false** (making it the correct answer to the "except" question).

◇ **Monocytes** and lymphocytes are classified as **agranular** (nongranular) **leukocytes** because they lack conspicuous

chemical granules in their cytoplasm.

◇ Granular leukocytes (granulocytes) include neutrophils, eosinophils, and basophils. All other options are true: red blood cells are biconcave discs with a typical 120-day lifespan, and platelets play a key role in clotting.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 69

Which one of the following is/are NOT CORRECT description about tissues.

- A. Tissues are formed by groups of closely related cells and their product known as intracellular matrix.
- B. Abundant microvilli which form prominent brush border on the luminal surface are the characteristic of epithelial tissues that transport substances or particles on the surface.
- C. Transitional epithelium is a type of epithelium whose appearance varies considerably depending on the degree of distension to which it is subjected.
- D. Non-keratinized stratified squamous epithelium is normally found in the body regions where the surface is dry or subject to severe mechanical abrasion.
- E. Epithelial tissue is characterized by abundant extracellular matrix.

Correct Answer: B, D, and E are all NOT CORRECT descriptions about tissues.

Explanation: Since the question asks which "is/are NOT CORRECT," multiple options contain fundamental biological errors:

✓ **B is NOT CORRECT:** Microvilli (forming the brush border) function to dramatically increase surface area for **absorption**. It is **cilia** that are motile and transport substances or particles along the epithelial surface.

✓ **D is NOT CORRECT:** Non-keratinized stratified squamous epithelium lines **moist** environments subject to friction (such as the oral cavity, esophagus, vagina, and anal canal). It is the **keratinized** stratified squamous epithelium that forms the **dry** outer protective barrier of the body (the epidermis) .

✓E is NOT CORRECT: Epithelial tissue is characterized by closely packed sheets of cells with **minimal or scarce extracellular matrix**. It is **connective tissue** that is characterized by an abundant extracellular matrix.

◇ *Note*: Options A and C are correct descriptions. Transitional epithelium indeed undergoes shape transitions depending on the degree of organ distension.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 70

Which of the following reasons makes histology dependent on the use of microscopes?

- A. The small size of cells and matrix components
- B. The small size of tissues
- C. The small size of organs
- D. All of the above
- E. None of the above

Correct Answer: A. The small size of cells and matrix components

Explanation: Histology is defined as the microscopic study of cells and extracellular matrix components. Because of **the extremely small size of individual cells and extracellular matrix macromolecules**, optical magnification (microscopy) is required to view and study how these components are arranged. Tissues and organs, by contrast, are macroscopic structures that can often be identified with the naked eye.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 71

Which of the following body tissues is most frequently subject to wear and tear due to its superficial location?

- A. Nervous tissue
- B. Muscle tissue
- C. Connective tissue
- D. Epithelial tissue
- E. All of the above

Correct Answer: D. Epithelial tissue

Explanation: Epithelial tissue forms the outer layer of the skin (the epidermis) and lines the internal cavities and lumina of organs. Because of its superficial location, it is directly exposed to the external environment and is the tissue most frequently subject to physical friction, abrasion, and general wear and tear.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 72

Which of the following descriptions about epithelial tissues is wrong?

- A. Epithelial tissues mostly exist on the free body surfaces or line body cavities and lumina of tubular organs.
- B. Epithelial tissues have diverse embryological origin from all germ layers.
- C. The apical, lateral and basal surfaces of epithelial cells reveal structural modifications based on the functional requirements of the tissue.
- D. Epithelial tissues are avascular.
- E. All epithelial tissues rest on a basal lamina.

Correct Answer: E. All epithelial tissues rest on a basal lamina.

Explanation: This statement is **wrong** (making it the correct choice). While the vast majority of epithelial tissues rest on a basal lamina (basement membrane), there are notable exceptions in the human body. For example, **ependymal cells** (the epithelial cells lining the ventricles of the brain and the central canal of the spinal cord) do not rest on a basal lamina; instead, their basal processes interdigitate directly with underlying astrocytes. All other options are anatomically correct:

✓ Epithelium lines free body surfaces, cavities, and lumina.

✓ Epithelia originate from all three embryonic germ layers.

✓ Apical, lateral, and basal surfaces exhibit functional modifications.

✓ Epithelia are notoriously avascular.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 73

In histology laboratory session, suppose you are given a histological slide containing all the four basic types of tissues. Which of the following histological features can help you to distinguish connective tissue from the other basic tissues?

- A. Highest cellularity
- B. Abundant cellular junctions connecting adjacent cells together
- C. Abundant intercellular substance separating the cells from each other
- D. The presence of blood vessels
- E. The presence of motor and sensory nerve endings

Correct Answer: C. Abundant intercellular substance separating the cells from each other

Explanation: Unlike epithelial, muscle, or nervous tissues—which are characterized by high cellularity and closely packed cells—**connective tissue** is distinguished by having relatively few cells suspended within an **abundant extracellular matrix (intercellular substance)** consisting of ground substance and protein fibers.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 74

Tendons, ligaments, retinacula, cartilages or bones are chiefly composed of _____

- a. Epithelial tissue
- b. Nervous tissues
- c. Muscle tissue
- d. Connective tissue
- e. All of the above

Correct Answer: d. Connective tissue

Explanation: Bones, tendons, ligaments, retinacula, and cartilages are all classified anatomically as supportive or dense **connective tissues**. They are composed of specialized cells (like osteocytes or chondrocytes) embedded in a fibrous extracellular matrix.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 75

The basic tissue types that constitute the brain and spinal cord is

- a. Connective tissue
- b. Skeletal tissue
- c. Blood tissue
- d. Nervous tissue
- e. Epithelium

Correct Answer: d. Nervous tissue

Explanation: The brain and spinal cord constitute the Central Nervous System (CNS). The CNS is composed almost exclusively of **nervous tissue**, which consists of structural and functional units called neurons and their supporting cells, the neuroglia (such as astrocytes, oligodendrocytes, and microglia).

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 76

Most tissue undergo constant cell turnover because of continuous cell division and ongoing death of cells. The turnover rate of cells is very rapid and high in which type of the following tissue?

- A. Connective tissue
- B. Muscles tissue
- C. Epithelial tissue
- D. Nervous tissue

Correct Answer: C. Epithelial tissue

Explanation: Epithelial tissues cover the free surfaces of the body and line its internal cavities. Because of this exposure, they are subject to constant physical friction, chemical exposure, and general wear and tear. To maintain structural integrity, epithelial tissues undergo rapid and continuous mitotic division to replace dead or sloughed cells, giving them the highest and most rapid turnover rate of the four basic tissue types.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 77

Each biological tissue consists of two interacting structural components known as _____ and _____.

Correct Answer: cells and extracellular matrix (ECM)

Explanation: By histological definition, every biological tissue in the human body is a cooperative mixture composed of two interacting structural components: the **living cells** themselves and the surrounding non-living **extracellular matrix (ECM)**. The cells produce and maintain the matrix, while the matrix provides mechanical/structural support and regulates cell behavior.

[24/09/2026 9:39 AM] Imaanaa Harmee: Question 78

Nerve tissues are principally composed two groups of cells: These are _____ and _____.

Correct Answer: **neurons** and **neuroglia** (or supporting cells)

Explanation: Nervous tissue is composed of two primary categories of cells:

◊ **Neurons:** The highly specialized structural and functional units of the nervous system responsible for receiving, integrating, and transmitting electrical impulses.

◊ **Neuroglia** (or glial cells): The non-neuronal supporting cells (such as astrocytes, oligodendrocytes, microglia, Schwann cells, and satellite cells) that nourish, insulate, protect, and physically support the neurons.